M21/MAB1 Correction material
UV-VIS optical spectroscopy

-Units I

a) One knows that $\lambda = c.T = \frac{1}{\sigma}$ i.e. $\sigma = \frac{1}{\lambda}$ or $\sigma(cm^{-1}).10^{-2} = \frac{1}{\lambda}$

If λ in nm then

$$10^{-2}.\sigma(cm^{-1}) = \frac{1}{10^{-9}.\lambda(nm)} = \frac{10^{7}}{10^{-2}.\lambda(nm)} \quad \text{i.e.} \quad \sigma(cm^{-1}) = \frac{10^{7}}{\lambda(nm)}$$

Remark: σ and E are linear but λ is inversely proportional !

$$\lambda = \frac{c}{\upsilon} = \frac{c}{f} \quad \text{or we have} \quad E = hf = h\upsilon \quad \text{then}$$

$$\lambda = \frac{c.h}{E} \text{ or } \lambda(nm).10^{-9} = \frac{c.h}{E(eV)*1.6\ 10^{-19}}$$

Numerical application : $\lambda(nm) = \frac{3.10^{8} \times 6.63\,10^{-34}}{E(eV) \times 1.6\,10^{-19} \times 10^{-9}} = 12.43\frac{10^{-7}}{E(eV)}$ then

$$\lambda(nm) = \frac{1243}{E(eV)}$$

b) Visible range from 350 to 750 nm

Numerical application :

$$E_{350} = \frac{1243}{350} = 3.55eV$$
$$E_{750} = \frac{1243}{750} = 1.66eV$$

c) One knows that $\lambda = \frac{c.h}{E}$ then $d\lambda = \Delta\lambda = -\frac{c.h.\Delta E}{E^{2}}$ or $\frac{d\lambda}{dE} = \frac{d}{dE}\left(\frac{c.h}{E}\right)$

then $\Delta\lambda = \frac{\lambda^{2}}{c.h}\Delta E(eV) \times e$

and therefore

$$\Delta\lambda(nm) = \frac{\lambda(nm)^{2} \times 10^{-9}}{c.h}\Delta E(eV) \times e = \frac{\lambda(nm)^{2} \times 10^{-9}}{3\,10^{8} \times 6.63\,10^{-34}}\Delta E(eV) \times 1.6\,10^{-9}$$

finally $$\Delta\lambda(nm) = \frac{\lambda(nm)^{2}\ \Delta E(eV)}{1243}$$

- **Units II**

a) The material is $Al_{0.3}Ga_{0.7}As$ then for x<0.45 we use E(x)=1.424+1.247x

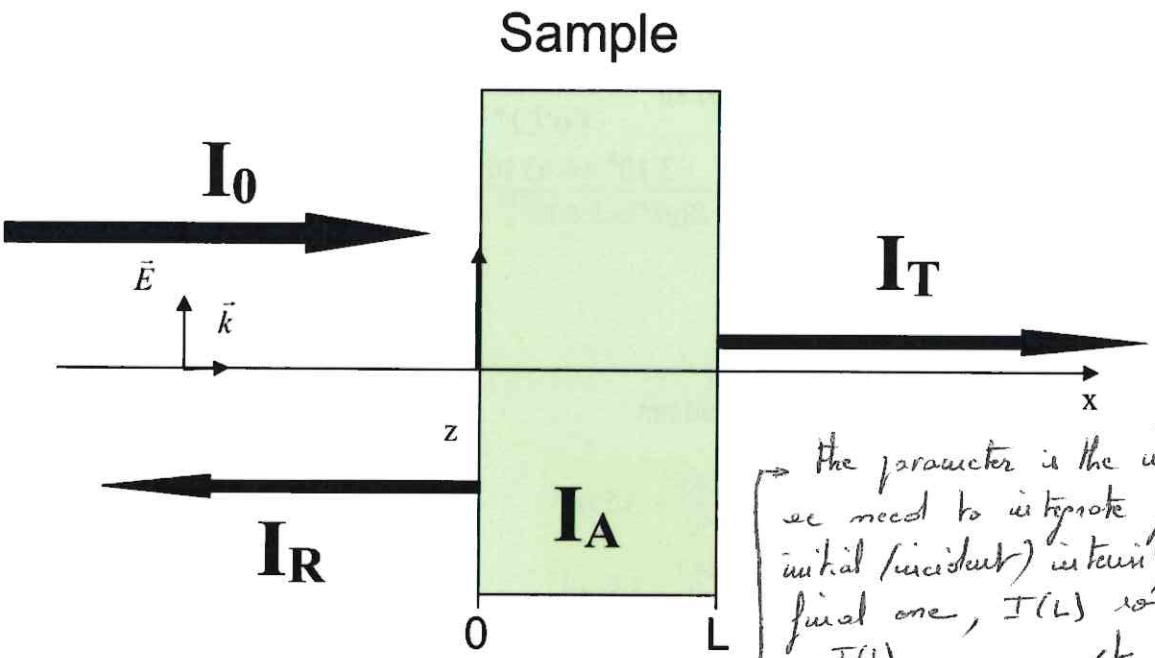
Numerical application: E(0.3)=1.424+0.3*1.247
<u>E(0.3)=1.798 eV=691.28 nm</u>

b) E(x) ϵ [1.424,1.424+1.247*0.45]

E(x) ϵ [1.424,1.985] eV [626.19, 872.89] nm

- **Absorption I**

a) *Loi de Beer-lambert



We have $\frac{dI}{dx} = -\alpha(\lambda_0).I(x)$

After integration $\int_0^L \frac{dI}{I} = -\alpha \int_0^L dx = -\alpha.L$

$I(L) = I_0.e^{-\alpha.L}$ Beer-Lambert-(Bouguer) Law

One can define the transmittance (transmission) as $T = \frac{I(L)}{I_0} = e^{-\alpha.L}$

the parameter is the intensity so we need to integrate from the initial (incident) intensity to the final one, I(L) so

$$\int_{I_0}^{I(L)} \frac{dI}{I} = -\alpha \int_0^L dx = -\alpha L$$

$$\ln(I(L)) - \ln I_0 = -\alpha L$$

(the primitive of $\frac{1}{x}$ is $\ln x$)

$$\ln(I(L)) = -\alpha L + \ln I_0$$

finally we apply the exponential function to both terms and we obtain:

$$I(L) = e^{-\alpha L} I_0 = I_0 e^{-\alpha L}$$

(reminder $e^{ab} = e^a + e^b$)

And one define the absorbance as $A = -\log_{10}\left(\frac{I(L)}{I_0}\right) = -\log_{10} T$

Precision: A is the absorption 1-R-T different from the absorbance

* Beer-Lambert demonstration using the electromagnetic field

Remarks: the bold letters represent vectors

Assuming that a planar electromagnetic wave propagates along x with E align along z the field is as follow:

$\mathbf{E} = \mathbf{E_0} \,.\, e^{i(\mathbf{k}.\mathbf{r}-\omega t)} = E_0\, \mathbf{e_z}\, e^{i(\mathbf{k}.\mathbf{r}-\omega t)}$

then $\mathbf{k.r} = k_0.x$ with $k_0 = \frac{2\pi\, n_0}{\lambda}$ where n_0 is the vacuum refractive index (1).

then at x = 0, $E = E_0\, e^{-i\omega t}$

and x = L , $E_t = E_0\, e^{i(k0.L-\omega t)}$

with in the material, a complex refractive index $n_c = n+i\chi$ with n_r the real part linked to the dispersion and χ the imaginary part linked to the absorption.

After replacing by the complex refractive index one gets $E_t = E0\, e^{i(2\pi n.L/\lambda+i2\pi\chi L/\lambda\, -\omega t)}$ or 3 terms including 2 phase terms $e^{i2\pi n.L/\lambda}$ and $e^{-i\omega t}$ and 1 attenuation term $e^{-2\pi\chi L/\lambda}$

Then knowing that the intensity is proportional to the square of the electromagnetic field:

$$\frac{I_t}{I_0} = \frac{K.\left|\vec{E}_t\right|^2}{K.\left|\vec{E}_0\right|^2} = e^{-\alpha L} \quad \text{and } \alpha = 4\pi\chi L/\lambda$$

And then one retrieve the Beer-Lambert law.

b) In Chemistry the molar concentration is c_m with a unit in $mol.L^{-1}$:

Then $$A = \frac{\alpha.L}{\ln 10} = \frac{c_m.\varepsilon_\lambda.L}{\ln 10}$$

One can then define the molar absorption (or extinction) coefficient ε_λ in $L.cm^{-1}.mol^{-1}$

c) When mixing different compounds, it is simple to demonstrate using the transmittance formula that the absorbance can be added (this is only through is the absorption process are independent from each other):

↳ the transmittance of a diluted compound is the transmittance of the solvent multiply by the transmittance of the compound

$T_{total} = T_{solvent} \times T_{compound} \Rightarrow \log_{10} T_{total} = \log_{10} T_{solvent} + \log_{10} T_{compound}$

$A_{total} = A_{solvent} + A_{compound}$ (reminder: $\ln(ab) = \ln a + \ln b$)
if we generalize ...

$$A = \sum_{i=1}^{n} A_i = \sum_{i=1}^{n} c'_m . \varepsilon'_\lambda . L$$

The method to determine the solvent contribution is to measure the solvent contribution first and then the absorbance of the solvent plus the compound:

Solvent only: $A=A_0$

Solvent + compound: $A=A_0+A_c$

Then $A-A_0=A_c$ gives the absorbance of the compound itself!

d) Example:

i- If T=0.5 @ 525 nm and L=10 mm

Numerical application: A=-log T =- log 0.5=0.3

and $\varepsilon_\lambda = \frac{A.\ln 10}{c_m.L}$ = (0.3*2.3)/(1.28.10^{-4} 1.10^{-1})=0.53 10^{7} cm^2/mol

(be careful with the unit).

ii- Now if one changes the concentration i.e. double the concentration then the absorbance is double.

$$A' = \frac{c'_m.\varepsilon_\lambda.L}{\ln 10} = 2A$$

AN : D'où A'=0.6 et T=10^{-2A}=0.25

-**Absorption II**

The absorption cross section, σ is given in cm^2

but α=σ.n where n is the atomic number per unit of volume in cm^{-3}

and α.L=σ.n.L=cm-3.cm2.cm=L^{-1}.cm^3.cm^{-1}.cm.mol.mol^{-1}
=(L^{-1}.mol)*(cm^3.mol^{-1}.cm^{-1})*cm
=(L^{-1}.mol)*(L.mol^{-1}.cm^{-1})* cm
= c_m * ε_λ * L

In fact they are talking about the same thing.

-**Absorption III**

a) The conservation of energy implies:

$$I_0 = I_R + I_A + I_T$$

where $$1 = \frac{I_R}{I_0} + \frac{I_A}{I_0} + \frac{I_T}{I_0} = R + a + T$$

One can define the reflectance (or better reflection coefficient in intensity).

The measured value is always T and then one finds a, through a=1-T where we see the complementarity between the absorbed and transmitted. In fact generally speaking one has also to consider the reflectance then a=1-(T+R). Then one obtains:

$$A = -\log(T - R)$$

b) Two possibilities for the experimental protocol:

- Either one substract the contribution of the refectance by measuring the transmission of the container with the solvent but without the compound to analyse.
- Or one measures T AND R but R is difficult to measure as one has to consider the angle.

c) If one considers as well that the light is scattered by the sample one will have another contribution to the losses one defines:

$1 = R + a + T'$ où $T' = T_{diff} + T_{dir}$

d) If finally, one is using a coherent source like a laser one will observed interferences as soon as the thickness because in the order of the wavelength. In this one cannot sum up the intensity contributions but the field contribution and then take the square of the sum of the field. In the case of incoherent source one can sum the square of each field i.e. each intensity.